STA ANALYSIS - SKY130 HD

Pre-Layout Static Timing Analysis of the Traffic Signal Controller (traffic_ctrl)

Roll No.: 24EG104E28

Execution

OpenSTA was run on my Ubuntu/UTM environment using the SKY130 HD timing libraries, on the Yosys-synthesized gate-level netlist of traffic_ctrl (29 flip-flops, about 150 gates, 1501.44 um2). All 16 corners were analyzed with a 50 ns clock (20 MHz). Clock uncertainty is 0.25 ns, clock transition 0.15 ns, and input/output delays are 20% of the period.

Important commands used

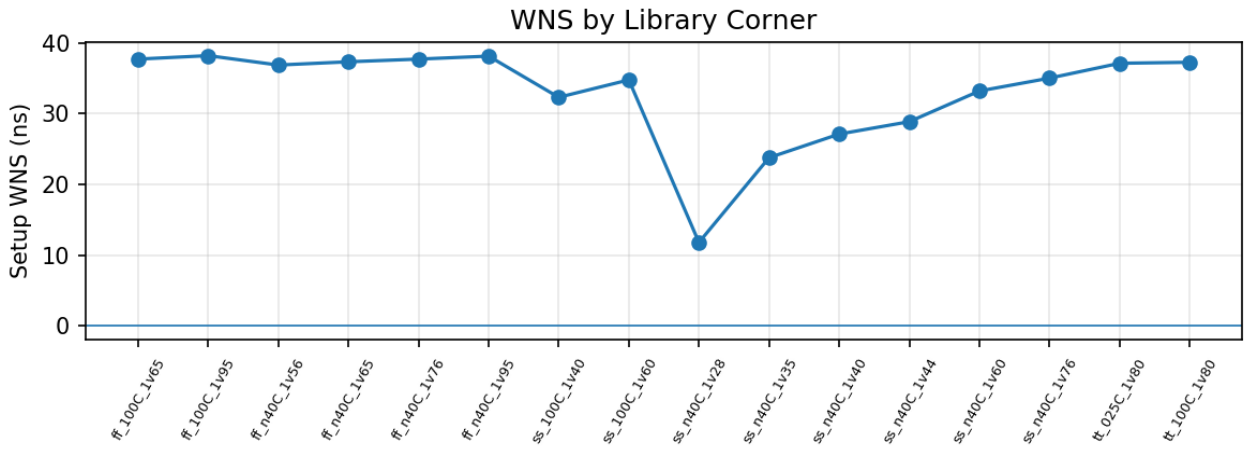
```
read_liberty <corner_library.lib>
read_verilog traffic_ctrl_netlist.v
link_design traffic_ctrl
read_sdc traffic_ctrl.sdc
report_checks -path_delay min_max -fields {nets cap slew input_pins fanout}
```

Note on clock period

At a first trial of 10 ns (100 MHz), 8 of 16 corners violated setup (worst -28.19 ns at ss_n40C_1v28), while hold was met everywhere. The controller reacts only to a slow tick, so the clock was relaxed to 50 ns, which meets timing in the corners listed below.

16-Corner Result Summary

#	Library corner	Process / Temp / V	Hold slack (ns)	Setup slack (ns)	Status
1	ff_100C_1v65	FF / +100 C / 1.65 V	0.12	37.73	MET
2	ff_100C_1v95	FF / +100 C / 1.95 V	0.04	38.22	MET
3	ff_n40C_1v56	FF / -40 C / 1.56 V	0.18	36.89	MET
4	ff_n40C_1v65	FF / -40 C / 1.65 V	0.13	37.35	MET
5	ff_n40C_1v76	FF / -40 C / 1.76 V	0.09	37.73	MET
6	ff_n40C_1v95	FF / -40 C / 1.95 V	0.04	38.15	MET
7	ss_100C_1v40	SS / +100 C / 1.40 V	0.99	32.33	MET
8	ss_100C_1v60	SS / +100 C / 1.60 V	0.64	34.79	MET
9	ss_n40C_1v28	SS / -40 C / 1.28 V	2.08	11.81	MET
10	ss_n40C_1v35	SS / -40 C / 1.35 V	1.50	23.79	MET
11	ss_n40C_1v40	SS / -40 C / 1.40 V	1.23	27.13	MET
12	ss_n40C_1v44	SS / -40 C / 1.44 V	1.06	28.88	MET
13	ss_n40C_1v60	SS / -40 C / 1.60 V	0.66	33.22	MET
14	ss_n40C_1v76	SS / -40 C / 1.76 V	0.44	35.04	MET
15	tt_025C_1v80	TT / +025 C / 1.80 V	0.20	37.14	MET
16	tt_100C_1v80	TT / +100 C / 1.80 V	0.20	37.28	MET



Negative setup slack indicates a setup timing violation for the reported worst max path.

ff_100C_1v65

Meaning: Fast NMOS / Fast PMOS; temperature +100 C; supply 1.65 V.

Result: hold slack 0.12 ns; setup slack 37.73 ns - **MET**.

```
worst slack 37.73
worst slack 0.12
Startpoint: _309_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _310_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _309_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.03    0.27    0.27 ^ _309_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           sync_p[0] (net)
                  0.03    0.00    0.27 ^ _172_/B_N (sky130_fd_sc_hd__nor2b_1)
                  0.06    0.08    0.35 ^ _172_/Y (sky130_fd_sc_hd__nor2b_1)
     1    0.00                           _001_ (net)
                  0.06    0.00    0.35 ^ _310_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.35   data arrival time
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _310_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.02    0.23   library hold time
                                  0.23   data required time
---------------------------------------------------------------------------
                                  0.23   data required time
                                 -0.35   data arrival time
---------------------------------------------------------------------------
                                  0.12   slack (MET)
Startpoint: _331_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: b_red (output port clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _331_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.15    0.36    0.36 ^ _331_/Q (sky130_fd_sc_hd__dfxtp_1)
     8    0.02                           state[2] (net)
                  0.15    0.00    0.36 ^ _159_/B_N (sky130_fd_sc_hd__nor2b_1)
                  0.13    0.16    0.52 ^ _159_/Y (sky130_fd_sc_hd__nor2b_1)
     3    0.01                           _142_ (net)
                  0.13    0.00    0.52 ^ _163_/C_N (sky130_fd_sc_hd__nor3b_1)
```

ff_100C_1v95

Meaning: Fast NMOS / Fast PMOS; temperature +100 C; supply 1.95 V.

Result: hold slack 0.04 ns; setup slack 38.22 ns - **MET**.

```
worst slack 38.22
worst slack 0.04
Startpoint: _309_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _310_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _309_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.03    0.21    0.21 ^ _309_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           sync_p[0] (net)
                  0.03    0.00    0.21 ^ _172_/B_N (sky130_fd_sc_hd__nor2b_1)
                  0.04    0.07    0.28 ^ _172_/Y (sky130_fd_sc_hd__nor2b_1)
     1    0.00                           _001_ (net)
                  0.04    0.00    0.28 ^ _310_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.28   data arrival time
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _310_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.01    0.24   library hold time
                                  0.24   data required time
---------------------------------------------------------------------------
```

```
                                  0.24   data required time
                                 -0.28   data arrival time
---------------------------------------------------------------------------
                                  0.04   slack (MET)
Startpoint: _331_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: b_red (output port clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _331_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.13    0.29    0.29 ^ _331_/Q (sky130_fd_sc_hd__dfxtp_1)
     8    0.02                           state[2] (net)
                  0.13    0.00    0.29 ^ _159_/B_N (sky130_fd_sc_hd__nor2b_1)
                  0.10    0.12    0.41 ^ _159_/Y (sky130_fd_sc_hd__nor2b_1)
     3    0.01                           _142_ (net)
                  0.10    0.00    0.41 ^ _163_/C_N (sky130_fd_sc_hd__nor3b_1)
```

ff_n40C_1v56

Meaning: Fast NMOS / Fast PMOS; temperature -40 C; supply 1.56 V.

Result: hold slack 0.18 ns; setup slack 36.89 ns - **MET**.

```
worst slack 36.89
worst slack 0.18
Startpoint: _309_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _310_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _309_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.03    0.32    0.32 ^ _309_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           sync_p[0] (net)
                  0.03    0.00    0.32 ^ _172_/B_N (sky130_fd_sc_hd__nor2b_1)
                  0.07    0.10    0.42 ^ _172_/Y (sky130_fd_sc_hd__nor2b_1)
     1    0.00                           _001_ (net)
                  0.07    0.00    0.42 ^ _310_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.42   data arrival time
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _310_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.02    0.23   library hold time
                                  0.23   data required time
---------------------------------------------------------------------------
                                  0.23   data required time
                                 -0.42   data arrival time
---------------------------------------------------------------------------
                                  0.18   slack (MET)
Startpoint: _331_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: b_red (output port clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _331_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.19    0.43    0.43 ^ _331_/Q (sky130_fd_sc_hd__dfxtp_1)
     8    0.02                           state[2] (net)
                  0.19    0.00    0.43 ^ _159_/B_N (sky130_fd_sc_hd__nor2b_1)
                  0.16    0.21    0.64 ^ _159_/Y (sky130_fd_sc_hd__nor2b_1)
     3    0.01                           _142_ (net)
                  0.16    0.00    0.64 ^ _163_/C_N (sky130_fd_sc_hd__nor3b_1)
```

ff_n40C_1v65

Meaning: Fast NMOS / Fast PMOS; temperature -40 C; supply 1.65 V.

Result: hold slack 0.13 ns; setup slack 37.35 ns - **MET**.

```
worst slack 37.35
worst slack 0.13
Startpoint: _309_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _310_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _309_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.03    0.28    0.28 ^ _309_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           sync_p[0] (net)
                  0.03    0.00    0.28 ^ _172_/B_N (sky130_fd_sc_hd__nor2b_1)
                  0.06    0.09    0.37 ^ _172_/Y (sky130_fd_sc_hd__nor2b_1)
     1    0.00                           _001_ (net)
                  0.06    0.00    0.37 ^ _310_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.37   data arrival time
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _310_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.01    0.24   library hold time
                                  0.24   data required time
---------------------------------------------------------------------------
```

```
                                  0.24   data required time
                                 -0.37   data arrival time
-----------------------------------------------------------------------------
                                  0.13   slack (MET)
Startpoint: _331_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: b_red (output port clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _331_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.17    0.38    0.38 ^ _331_/Q (sky130_fd_sc_hd__dfxtp_1)
     8    0.02                           state[2] (net)
                  0.17    0.00    0.38 ^ _159_/B_N (sky130_fd_sc_hd__nor2b_1)
                  0.14    0.18    0.57 ^ _159_/Y (sky130_fd_sc_hd__nor2b_1)
     3    0.01                           _142_ (net)
                  0.14    0.00    0.57 ^ _163_/C_N (sky130_fd_sc_hd__nor3b_1)
```

ff_n40C_1v76

Meaning: Fast NMOS / Fast PMOS; temperature -40 C; supply 1.76 V.

Result: hold slack 0.09 ns; setup slack 37.73 ns - **MET**.

```
worst slack 37.73
worst slack 0.09
Startpoint: _309_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _310_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _309_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.03    0.25    0.25 ^ _309_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           sync_p[0] (net)
                  0.03    0.00    0.25 ^ _172_/B_N (sky130_fd_sc_hd__nor2b_1)
                  0.05    0.08    0.33 ^ _172_/Y (sky130_fd_sc_hd__nor2b_1)
     1    0.00                           _001_ (net)
                  0.05    0.00    0.33 ^ _310_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.33   data arrival time
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _310_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.01    0.24   library hold time
                                  0.24   data required time
---------------------------------------------------------------------------
                                  0.24   data required time
                                 -0.33   data arrival time
---------------------------------------------------------------------------
                                  0.09   slack (MET)
Startpoint: _331_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: b_red (output port clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _331_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.15    0.34    0.34 ^ _331_/Q (sky130_fd_sc_hd__dfxtp_1)
     8    0.02                           state[2] (net)
                  0.15    0.00    0.34 ^ _159_/B_N (sky130_fd_sc_hd__nor2b_1)
                  0.12    0.16    0.50 ^ _159_/Y (sky130_fd_sc_hd__nor2b_1)
     3    0.01                           _142_ (net)
                  0.12    0.00    0.50 ^ _163_/C_N (sky130_fd_sc_hd__nor3b_1)
```

ff_n40C_1v95

Meaning: Fast NMOS / Fast PMOS; temperature -40 C; supply 1.95 V.

Result: hold slack 0.04 ns; setup slack 38.15 ns - **MET**.

```
worst slack 38.15
worst slack 0.04
Startpoint: _309_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _310_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _309_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.02    0.21    0.21 ^ _309_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           sync_p[0] (net)
                  0.02    0.00    0.21 ^ _172_/B_N (sky130_fd_sc_hd__nor2b_1)
                  0.05    0.07    0.27 ^ _172_/Y (sky130_fd_sc_hd__nor2b_1)
     1    0.00                           _001_ (net)
                  0.05    0.00    0.27 ^ _310_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.27   data arrival time
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _310_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.02    0.23   library hold time
                                  0.23   data required time
---------------------------------------------------------------------------
```

```
                                  0.23   data required time
                                 -0.27   data arrival time
---------------------------------------------------------------------------
                                  0.04   slack (MET)
Startpoint: _331_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: b_red (output port clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _331_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.14    0.29    0.29 ^ _331_/Q (sky130_fd_sc_hd__dfxtp_1)
     8    0.02                           state[2] (net)
                  0.14    0.00    0.29 ^ _159_/B_N (sky130_fd_sc_hd__nor2b_1)
                  0.11    0.13    0.41 ^ _159_/Y (sky130_fd_sc_hd__nor2b_1)
     3    0.01                           _142_ (net)
                  0.11    0.00    0.41 ^ _163_/C_N (sky130_fd_sc_hd__nor3b_1)
```

ss_100C_1v40

Meaning: Slow NMOS / Slow PMOS; temperature +100 C; supply 1.40 V.

Result: hold slack 0.99 ns; setup slack 32.33 ns - **MET**.

```
worst slack 32.33
worst slack 0.99
Startpoint: _326_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _326_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _326_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.11    0.89    0.89 v _326_/Q (sky130_fd_sc_hd__dfxtp_1)
     3    0.01                           min_cnt[7] (net)
                  0.11    0.00    0.89 v _151_/A (sky130_fd_sc_hd__clkinv_1)
                  0.09    0.13    1.02 ^ _151_/Y (sky130_fd_sc_hd__clkinv_1)
     3    0.01                           _134_ (net)
                  0.09    0.00    1.02 ^ _294_/A (sky130_fd_sc_hd__nor3_1)
                  0.06    0.10    1.13 v _294_/Y (sky130_fd_sc_hd__nor3_1)
     1    0.00                           _017_ (net)
                  0.06    0.00    1.13 v _326_/D (sky130_fd_sc_hd__dfxtp_1)
                                  1.13   data arrival time
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _326_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.11    0.14   library hold time
                                  0.14   data required time
---------------------------------------------------------------------------
                                  0.14   data required time
                                 -1.13   data arrival time
---------------------------------------------------------------------------
                                  0.99   slack (MET)
Startpoint: _331_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: b_red (output port clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _331_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.36    1.16    1.16 ^ _331_/Q (sky130_fd_sc_hd__dfxtp_1)
     8    0.02                           state[2] (net)
                  0.36    0.00    1.16 ^ _159_/B_N (sky130_fd_sc_hd__nor2b_1)
```

ss_100C_1v60

Meaning: Slow NMOS / Slow PMOS; temperature +100 C; supply 1.60 V.

Result: hold slack 0.64 ns; setup slack 34.79 ns - **MET**.

```
worst slack 34.79
worst slack 0.64
Startpoint: _309_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _310_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _309_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.05    0.64    0.64 ^ _309_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           sync_p[0] (net)
                  0.05    0.00    0.64 ^ _172_/B_N (sky130_fd_sc_hd__nor2b_1)
                  0.11    0.18    0.82 ^ _172_/Y (sky130_fd_sc_hd__nor2b_1)
     1    0.00                           _001_ (net)
                  0.11    0.00    0.82 ^ _310_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.82   data arrival time
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _310_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.07    0.18   library hold time
                                  0.18   data required time
---------------------------------------------------------------------------
```

```
                                   0.18   data required time
                                  -0.82   data arrival time
---------------------------------------------------------------------------
                                   0.64   slack (MET)
Startpoint: _331_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: b_red (output port clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _331_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.28    0.82    0.82 ^ _331_/Q (sky130_fd_sc_hd__dfxtp_1)
     8    0.02                           state[2] (net)
                  0.28    0.00    0.82 ^ _159_/B_N (sky130_fd_sc_hd__nor2b_1)
                  0.25    0.37    1.19 ^ _159_/Y (sky130_fd_sc_hd__nor2b_1)
     3    0.01                           _142_ (net)
                  0.25    0.00    1.19 ^ _163_/C_N (sky130_fd_sc_hd__nor3b_1)
```

ss_n40C_1v28

Meaning: Slow NMOS / Slow PMOS; temperature -40 C; supply 1.28 V.

Result: hold slack 2.08 ns; setup slack 11.81 ns - **MET**.

```
worst slack 11.81
worst slack 2.08
Startpoint: _322_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _322_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _322_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.12    1.60    1.60 v _322_/Q (sky130_fd_sc_hd__dfxtp_1)
     3    0.00                           min_cnt[3] (net)
                  0.12    0.00    1.60 v _276_/A (sky130_fd_sc_hd__nand2_1)
                  0.19    0.25    1.85 ^ _276_/Y (sky130_fd_sc_hd__nand2_1)
     1    0.00                           _116_ (net)
                  0.19    0.00    1.85 ^ _280_/A1 (sky130_fd_sc_hd__a21oi_1)
                  0.05    0.19    2.04 v _280_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _013_ (net)
                  0.05    0.00    2.04 v _322_/D (sky130_fd_sc_hd__dfxtp_1)
                                  2.04   data arrival time
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _322_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.29   -0.04   library hold time
                                 -0.04   data required time
---------------------------------------------------------------------------
                                 -0.04   data required time
                                 -2.04   data arrival time
---------------------------------------------------------------------------
                                  2.08   slack (MET)
Startpoint: _331_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _320_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _331_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.85    2.69    2.69 ^ _331_/Q (sky130_fd_sc_hd__dfxtp_1)
     8    0.01                           state[2] (net)
                  0.85    0.00    2.69 ^ _159_/B_N (sky130_fd_sc_hd__nor2b_1)
```

ss_n40C_1v35

Meaning: Slow NMOS / Slow PMOS; temperature -40 C; supply 1.35 V.

Result: hold slack 1.50 ns; setup slack 23.79 ns - **MET**.

```
worst slack 23.79
worst slack 1.50
Startpoint: _322_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _322_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _322_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.09    1.21    1.21 v _322_/Q (sky130_fd_sc_hd__dfxtp_1)
     3    0.00                           min_cnt[3] (net)
                  0.09    0.00    1.21 v _276_/A (sky130_fd_sc_hd__nand2_1)
                  0.14    0.19    1.40 ^ _276_/Y (sky130_fd_sc_hd__nand2_1)
     1    0.00                           _116_ (net)
                  0.14    0.00    1.40 ^ _280_/A1 (sky130_fd_sc_hd__a21oi_1)
                  0.04    0.15    1.55 v _280_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _013_ (net)
                  0.04    0.00    1.55 v _322_/D (sky130_fd_sc_hd__dfxtp_1)
                                  1.55   data arrival time
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _322_/CLK (sky130_fd_sc_hd__dfxtp_1)
```

```
                        -0.20    0.05   library hold time
                                 0.05   data required time
-----------------------------------------------------------------------------
                                 0.05   data required time
                                -1.55   data arrival time
-----------------------------------------------------------------------------
                                 1.50   slack (MET)
Startpoint: _331_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: b_red (output port clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _331_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.65    1.92    1.92 ^ _331_/Q (sky130_fd_sc_hd__dfxtp_1)
     8    0.02                           state[2] (net)
                  0.65    0.00    1.92 ^ _159_/B_N (sky130_fd_sc_hd__nor2b_1)
```

ss_n40C_1v40

Meaning: Slow NMOS / Slow PMOS; temperature -40 C; supply 1.40 V.

Result: hold slack 1.23 ns; setup slack 27.13 ns - **MET**.

```
worst slack 27.13
worst slack 1.23
Startpoint: _322_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _322_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _322_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.08    1.03    1.03 v _322_/Q (sky130_fd_sc_hd__dfxtp_1)
     3    0.01                           min_cnt[3] (net)
                  0.08    0.00    1.03 v _276_/A (sky130_fd_sc_hd__nand2_1)
                  0.12    0.16    1.19 ^ _276_/Y (sky130_fd_sc_hd__nand2_1)
     1    0.00                           _116_ (net)
                  0.12    0.00    1.19 ^ _280_/A1 (sky130_fd_sc_hd__a21oi_1)
                  0.05    0.13    1.32 v _280_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _013_ (net)
                  0.05    0.00    1.32 v _322_/D (sky130_fd_sc_hd__dfxtp_1)
                                  1.32   data arrival time
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _322_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.16    0.09   library hold time
                                  0.09   data required time
---------------------------------------------------------------------------
                                  0.09   data required time
                                 -1.32   data arrival time
---------------------------------------------------------------------------
                                  1.23   slack (MET)
Startpoint: _331_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: b_red (output port clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _331_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.55    1.58    1.58 ^ _331_/Q (sky130_fd_sc_hd__dfxtp_1)
     8    0.02                           state[2] (net)
                  0.55    0.00    1.58 ^ _159_/B_N (sky130_fd_sc_hd__nor2b_1)
```

ss_n40C_1v44

Meaning: Slow NMOS / Slow PMOS; temperature -40 C; supply 1.44 V.

Result: hold slack 1.06 ns; setup slack 28.88 ns - **MET**.

```
worst slack 28.88
worst slack 1.06
Startpoint: _322_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _322_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _322_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.08    0.92    0.92 v _322_/Q (sky130_fd_sc_hd__dfxtp_1)
     3    0.01                           min_cnt[3] (net)
                  0.08    0.00    0.92 v _276_/A (sky130_fd_sc_hd__nand2_1)
                  0.10    0.14    1.06 ^ _276_/Y (sky130_fd_sc_hd__nand2_1)
     1    0.00                           _116_ (net)
                  0.10    0.00    1.06 ^ _280_/A1 (sky130_fd_sc_hd__a21oi_1)
                  0.04    0.11    1.18 v _280_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _013_ (net)
                  0.04    0.00    1.18 v _322_/D (sky130_fd_sc_hd__dfxtp_1)
                                  1.18   data arrival time
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _322_/CLK (sky130_fd_sc_hd__dfxtp_1)
```

```
                          -0.13    0.12   library hold time
                                   0.12   data required time
-----------------------------------------------------------------------------
                                   0.12   data required time
                                  -1.18   data arrival time
-----------------------------------------------------------------------------
                                   1.06   slack (MET)
Startpoint: _331_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: b_red (output port clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _331_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.48    1.38    1.38 ^ _331_/Q (sky130_fd_sc_hd__dfxtp_1)
     8    0.02                           state[2] (net)
                  0.48    0.00    1.38 ^ _159_/B_N (sky130_fd_sc_hd__nor2b_1)
```

ss_n40C_1v60

Meaning: Slow NMOS / Slow PMOS; temperature -40 C; supply 1.60 V.

Result: hold slack 0.66 ns; setup slack 33.22 ns - **MET**.

```
worst slack 33.22
worst slack 0.66
Startpoint: _322_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _322_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _322_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.06    0.65    0.65 v _322_/Q (sky130_fd_sc_hd__dfxtp_1)
     3    0.01                           min_cnt[3] (net)
                  0.06    0.00    0.65 v _276_/A (sky130_fd_sc_hd__nand2_1)
                  0.07    0.10    0.75 ^ _276_/Y (sky130_fd_sc_hd__nand2_1)
     1    0.00                           _116_ (net)
                  0.07    0.00    0.75 ^ _280_/A1 (sky130_fd_sc_hd__a21oi_1)
                  0.04    0.08    0.82 v _280_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _013_ (net)
                  0.04    0.00    0.82 v _322_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.82   data arrival time
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _322_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.08    0.17   library hold time
                                  0.17   data required time
---------------------------------------------------------------------------
                                  0.17   data required time
                                 -0.82   data arrival time
---------------------------------------------------------------------------
                                  0.66   slack (MET)
Startpoint: _331_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: b_red (output port clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _331_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.34    0.91    0.91 ^ _331_/Q (sky130_fd_sc_hd__dfxtp_1)
     8    0.02                           state[2] (net)
                  0.34    0.00    0.91 ^ _159_/B_N (sky130_fd_sc_hd__nor2b_1)
```

ss_n40C_1v76

Meaning: Slow NMOS / Slow PMOS; temperature -40 C; supply 1.76 V.

Result: hold slack 0.44 ns; setup slack 35.04 ns - **MET**.

```
worst slack 35.04
worst slack 0.44
Startpoint: _322_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _322_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _322_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.05    0.50    0.50 v _322_/Q (sky130_fd_sc_hd__dfxtp_1)
     3    0.01                           min_cnt[3] (net)
                  0.05    0.00    0.50 v _276_/A (sky130_fd_sc_hd__nand2_1)
                  0.05    0.08    0.58 ^ _276_/Y (sky130_fd_sc_hd__nand2_1)
     1    0.00                           _116_ (net)
                  0.05    0.00    0.58 ^ _280_/A1 (sky130_fd_sc_hd__a21oi_1)
                  0.04    0.06    0.64 v _280_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _013_ (net)
                  0.04    0.00    0.64 v _322_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.64   data arrival time
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _322_/CLK (sky130_fd_sc_hd__dfxtp_1)
```

```
                          -0.05    0.20   library hold time
                                   0.20   data required time
-----------------------------------------------------------------------------
                                   0.20   data required time
                                  -0.64   data arrival time
-----------------------------------------------------------------------------
                                   0.44   slack (MET)
Startpoint: _331_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: b_red (output port clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _331_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.26    0.68    0.68 ^ _331_/Q (sky130_fd_sc_hd__dfxtp_1)
     8    0.02                           state[2] (net)
                  0.26    0.00    0.68 ^ _159_/B_N (sky130_fd_sc_hd__nor2b_1)
```

tt_025C_1v80

Meaning: Typical NMOS / Typical PMOS; temperature +025 C; supply 1.80 V.

Result: hold slack 0.20 ns; setup slack 37.14 ns - **MET**.

```
worst slack 37.14
worst slack 0.20
Startpoint: _309_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _310_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _309_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.03    0.33    0.33 ^ _309_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           sync_p[0] (net)
                  0.03    0.00    0.33 ^ _172_/B_N (sky130_fd_sc_hd__nor2b_1)
                  0.06    0.10    0.43 ^ _172_/Y (sky130_fd_sc_hd__nor2b_1)
     1    0.00                           _001_ (net)
                  0.06    0.00    0.43 ^ _310_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.43   data arrival time
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _310_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.02    0.23   library hold time
                                  0.23   data required time
---------------------------------------------------------------------------
                                  0.23   data required time
                                 -0.43   data arrival time
---------------------------------------------------------------------------
                                  0.20   slack (MET)
Startpoint: _331_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: b_red (output port clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _331_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.18    0.44    0.44 ^ _331_/Q (sky130_fd_sc_hd__dfxtp_1)
     8    0.02                           state[2] (net)
                  0.18    0.00    0.44 ^ _159_/B_N (sky130_fd_sc_hd__nor2b_1)
                  0.15    0.20    0.63 ^ _159_/Y (sky130_fd_sc_hd__nor2b_1)
     3    0.01                           _142_ (net)
                  0.15    0.00    0.63 ^ _163_/C_N (sky130_fd_sc_hd__nor3b_1)
```

tt_100C_1v80

Meaning: Typical NMOS / Typical PMOS; temperature +100 C; supply 1.80 V.

Result: hold slack 0.20 ns; setup slack 37.28 ns - **MET**.

```
worst slack 37.28
worst slack 0.20
Startpoint: _309_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _310_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _309_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.03    0.33    0.33 ^ _309_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           sync_p[0] (net)
                  0.03    0.00    0.33 ^ _172_/B_N (sky130_fd_sc_hd__nor2b_1)
                  0.06    0.10    0.42 ^ _172_/Y (sky130_fd_sc_hd__nor2b_1)
     1    0.00                           _001_ (net)
                  0.06    0.00    0.42 ^ _310_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.42   data arrival time
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.25    0.25   clock uncertainty
                          0.00    0.25   clock reconvergence pessimism
                                  0.25 ^ _310_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.02    0.23   library hold time
                                  0.23   data required time
---------------------------------------------------------------------------
```

```
                                  0.23   data required time
                                 -0.42   data arrival time
---------------------------------------------------------------------------
                                  0.20   slack (MET)
Startpoint: _331_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: b_red (output port clocked by clk)
Path Group: clk
Path Type: max
Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.15    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.15    0.00    0.00 ^ _331_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.17    0.43    0.43 ^ _331_/Q (sky130_fd_sc_hd__dfxtp_1)
     8    0.02                           state[2] (net)
                  0.17    0.00    0.43 ^ _159_/B_N (sky130_fd_sc_hd__nor2b_1)
                  0.14    0.19    0.62 ^ _159_/Y (sky130_fd_sc_hd__nor2b_1)
     3    0.01                           _142_ (net)
                  0.14    0.00    0.62 ^ _163_/C_N (sky130_fd_sc_hd__nor3b_1)
```

Conclusion

All 16 SKY130 HD corners were analyzed at a 50 ns clock. Worst setup slack is 11.81 ns (ss_n40C_1v28); worst hold slack is 0.04 ns (ff_100C_1v95). Timing is met in all 16 corners.